

TRANSIENT IMAGING

Computational Imaging – Spring 2024

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Slide credits

Some of these slides are based on the following content:

- [Computational Photography \(15-463\)](#). Carnegie Mellon University.

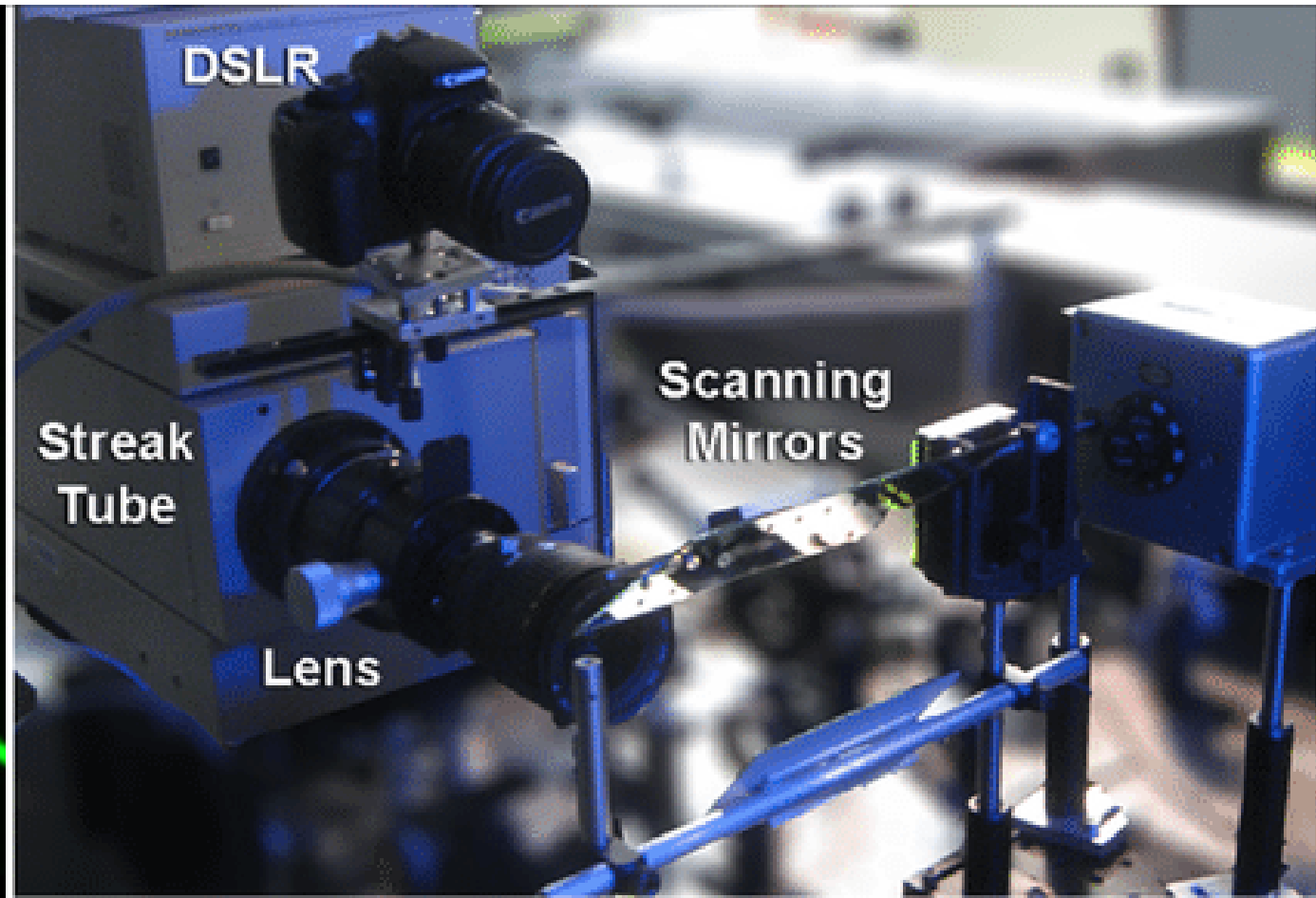
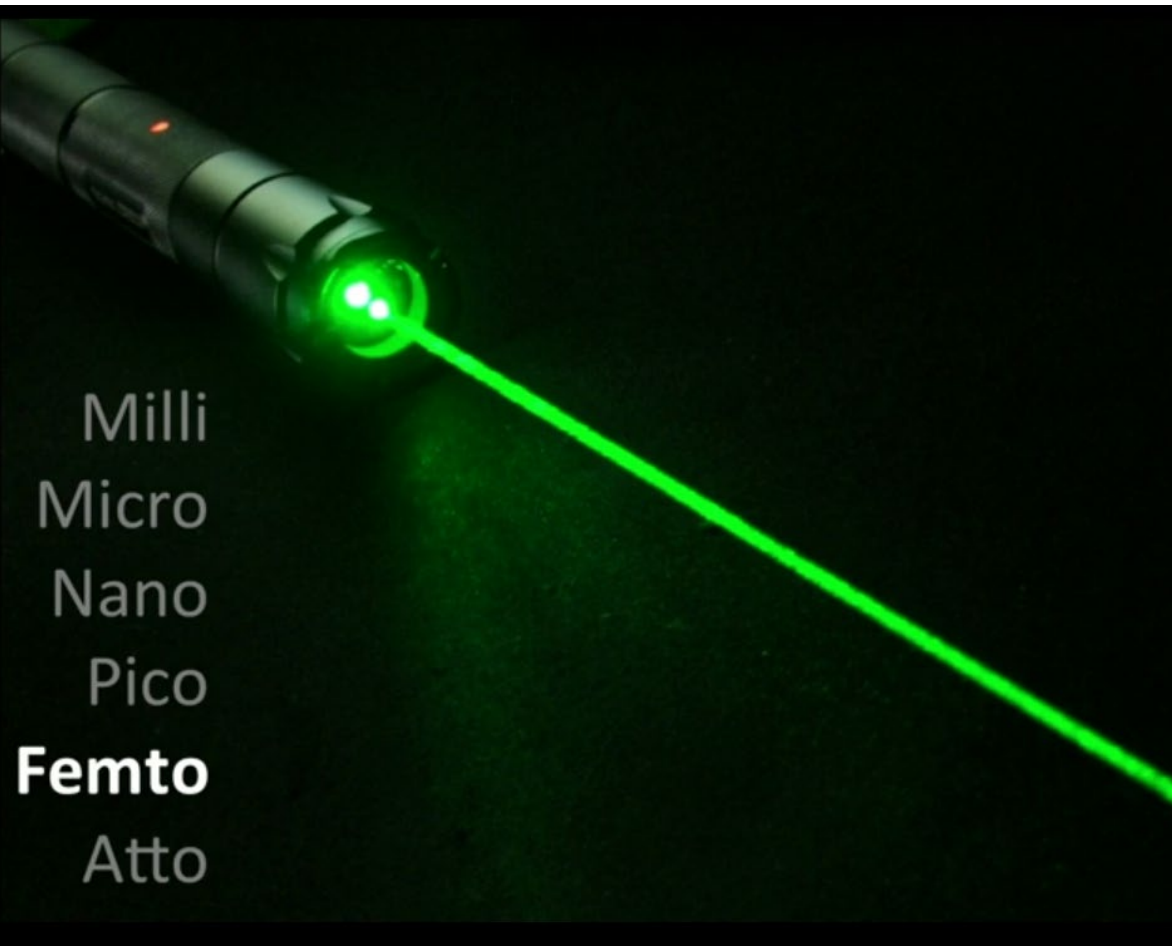
Transient imaging: Motivation

Information hidden in the **temporal domain of light propagation**:

- Capturing and visualizing light in motion → **Femtophotography (today)**
- Geometry reconstruction → **Correlation time-of-flight cameras (today)**
- Reflectance and media properties
- Non-line-of-sight imaging using SPAD sensors
 - Seeing hidden objects (next week)
 - Seeing through scattering media

Femto-cameras, SPAD sensors

Accurate but... expensive, difficult to operate, slow



Correlation-based Time-of-Flight Imaging

Correlation-based Time-of-Flight (ToF) cameras

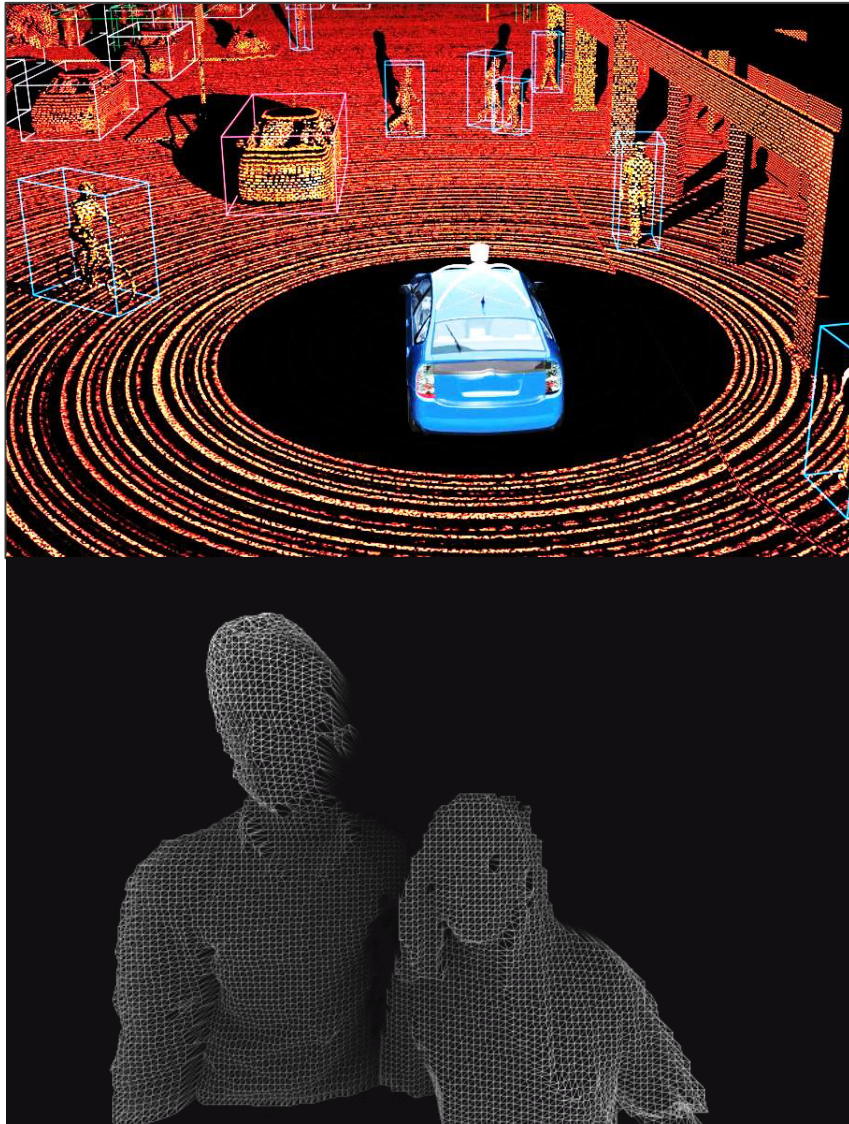
Affordable, off-the-shelf, real-time



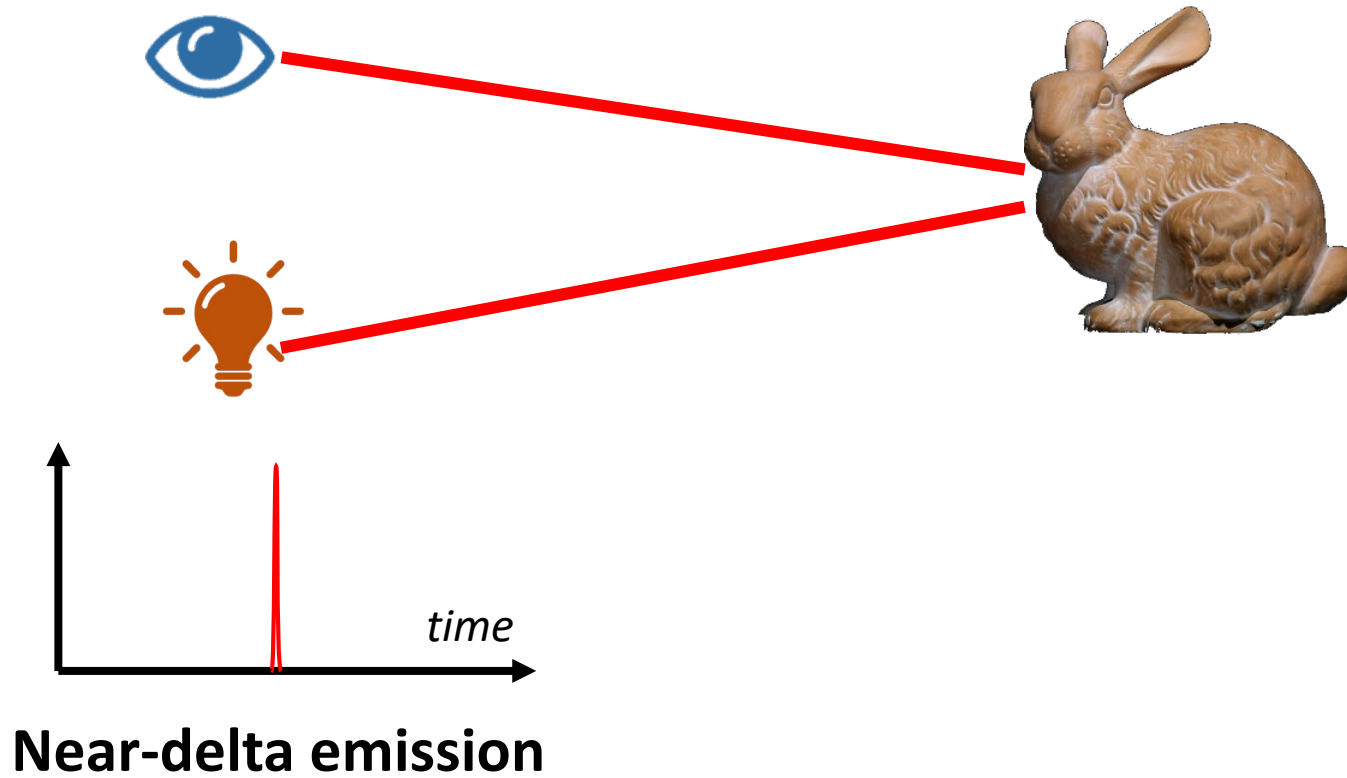
2009

Today

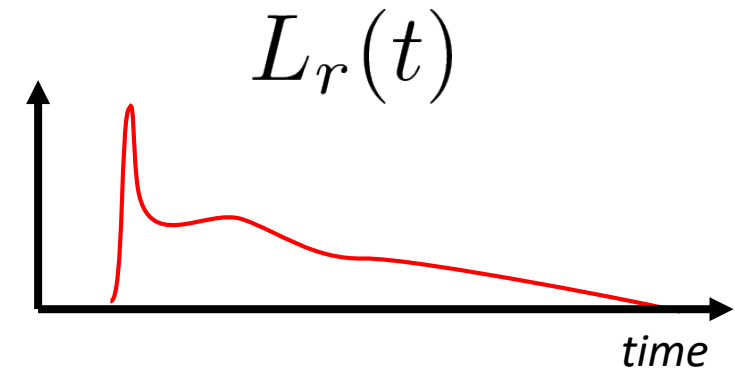
ToF applications: Real-time depth sensing



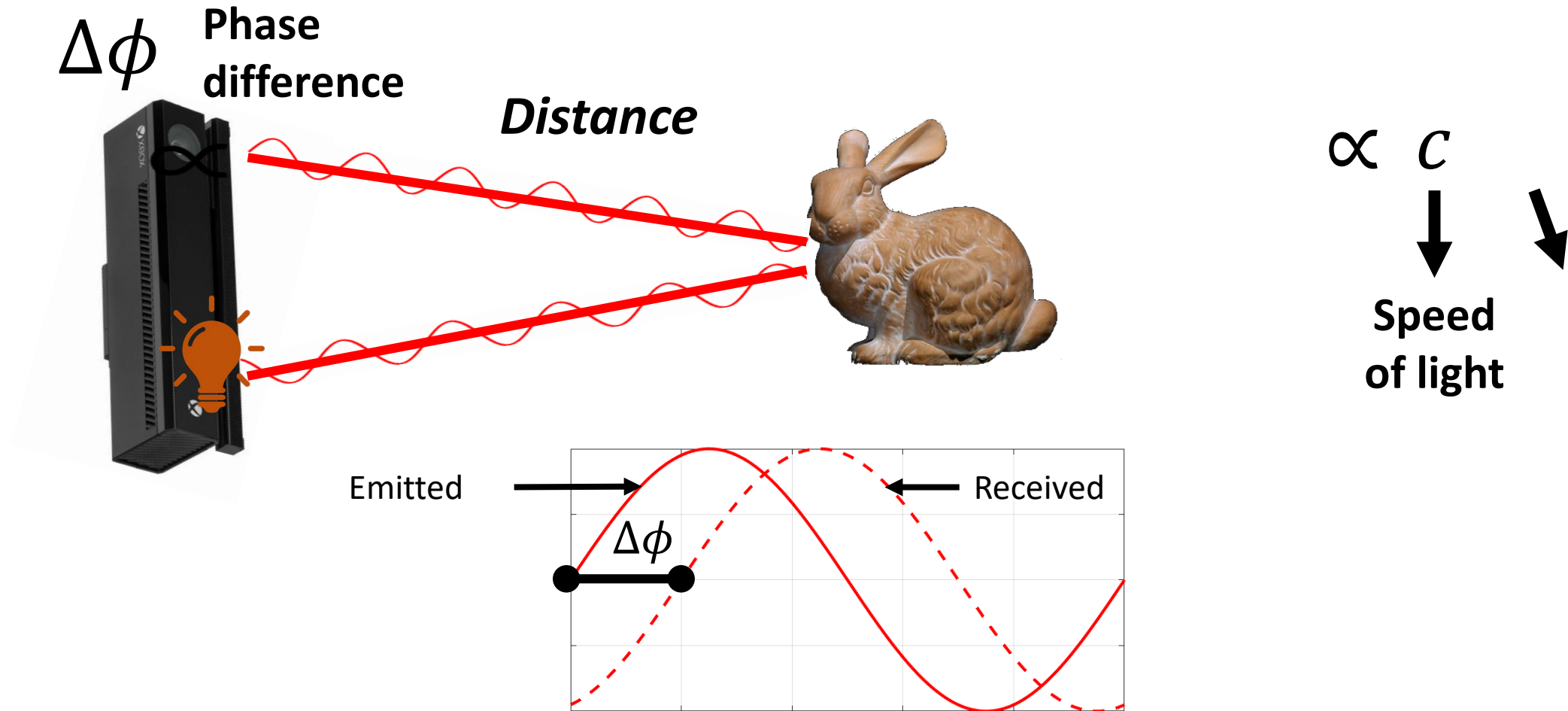
Impulse response function



Impulse response function



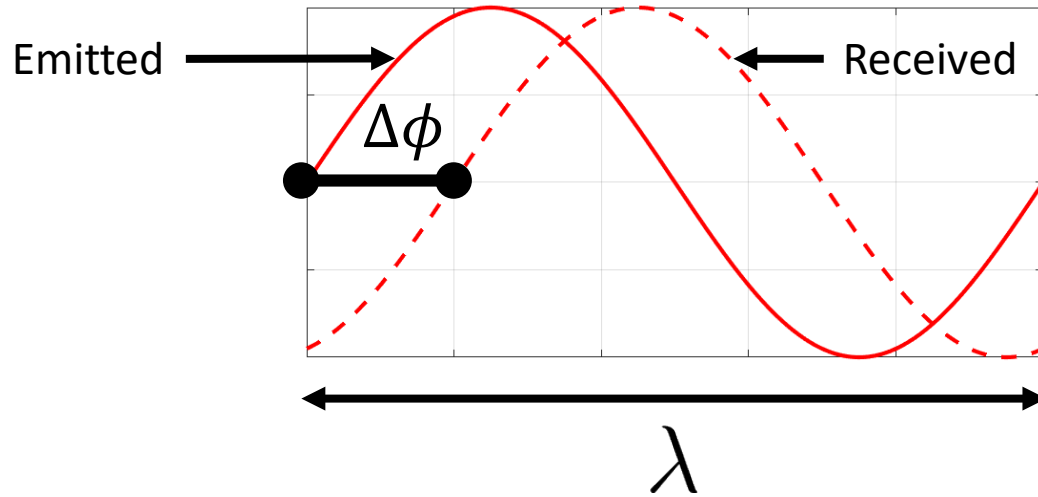
ToF cameras: Operational principles



ToF cameras: Operational principles

Exposure time $\gg$ modulation wavelength $\lambda = \frac{1}{f_\omega}$

$$\begin{array}{cc} T_e & \gg & \lambda \\ \text{(ms)} & & \text{(ns)} \end{array}$$

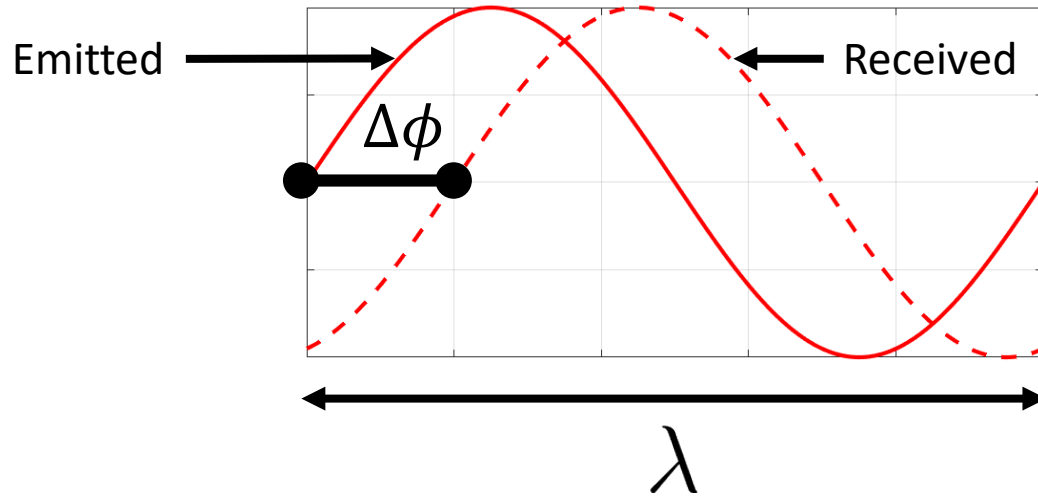


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**Cannot measure
 $\Delta\phi$ directly**

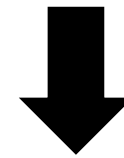


ToF cameras: Operational principles

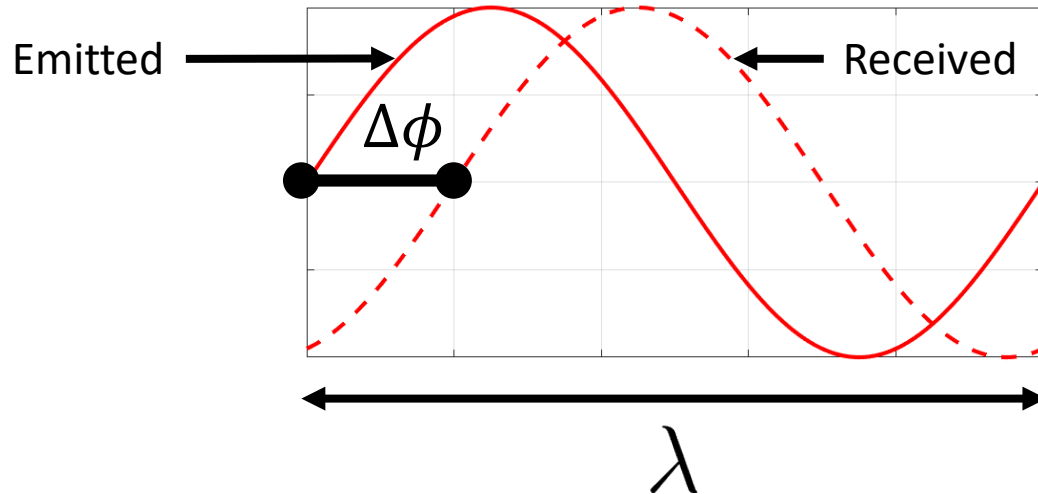
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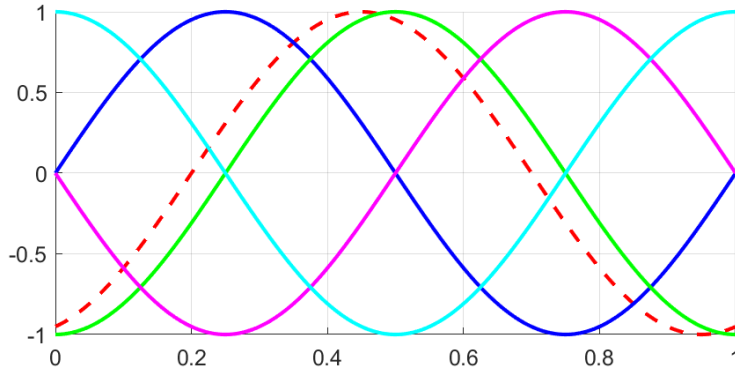


**Measure phase shift $\Delta\phi$
by CORRELATION**



Time-of-Flight cameras: Depth by correlation

- Correlation-based depth computation: Arctangent method

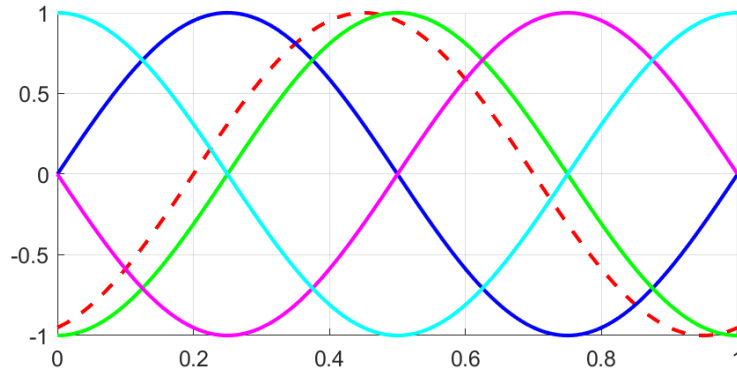


Received light
(at a sensor pixel)

$$L_r(t)$$

Time-of-Flight cameras: Depth by correlation

- Correlation-based depth computation: Arctangent method



Received light
(at a sensor pixel)

$$L_r(t)$$

Exposure functions

$$E_i(t) = \sin(2\pi f_\omega t - \phi_i)$$

— $\phi_1 = 0$

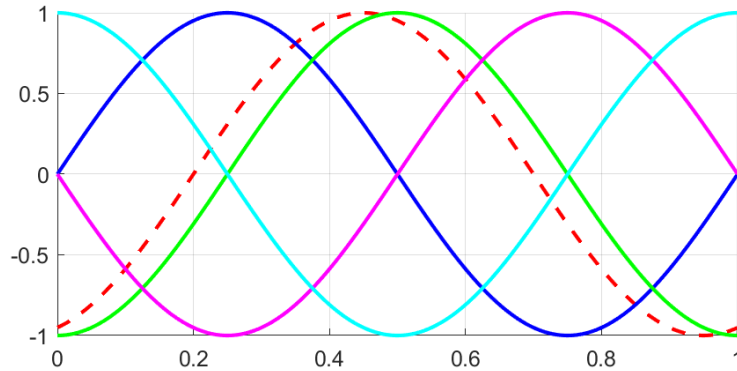
— $\phi_2 = \pi/2$

— $\phi_3 = \pi$

— $\phi_4 = 3\pi/2$

Time-of-Flight cameras: Depth by correlation

- Correlation-based depth computation: Arctangent method



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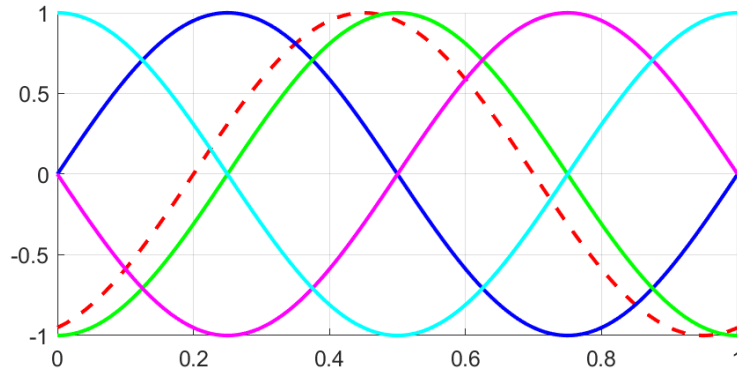
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Phase images:

$$I_i = \int_{T_e} [E_i(t) * L_r(t)] dt$$

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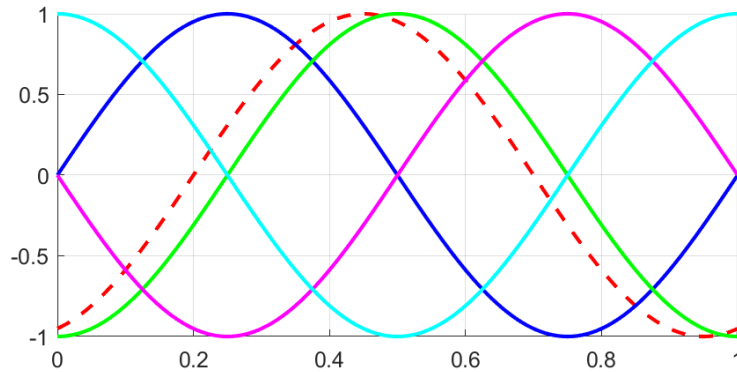
$$I_i = \int_{T_e} [E_i(t) * L_r(t)] dt$$

Phase shift:

$$\Delta\phi = \arctan\left(\frac{I_2 - I_4}{I_3 - I_1}\right)$$

Time-of-Flight cameras: Depth by correlation

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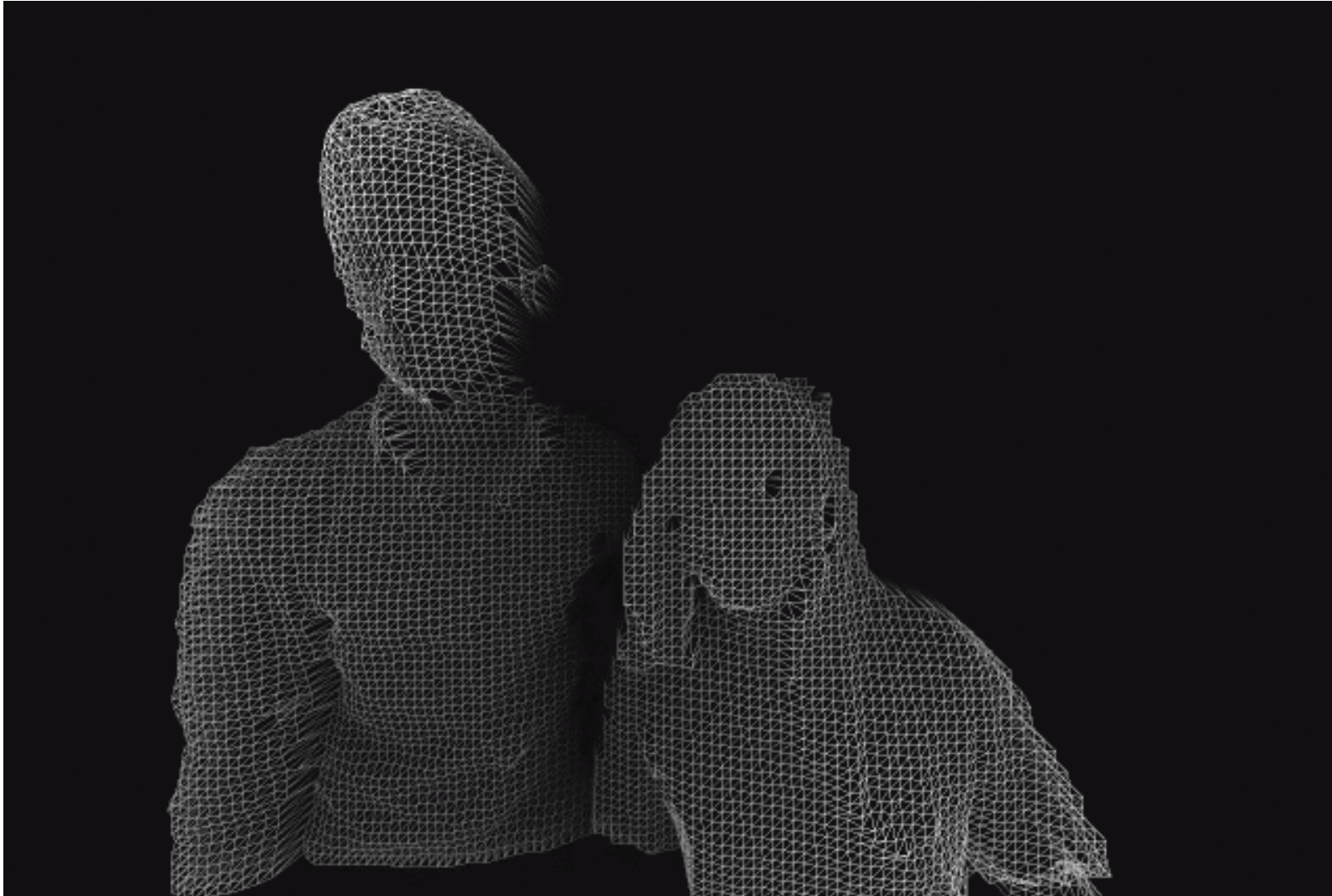
Phase shift:

$$\Delta\phi = \arctan\left(\frac{I_2 - I_4}{I_3 - I_1}\right)$$

Depth:

$$D = \frac{c \cdot \Delta\phi}{4\pi f_\omega}$$

Time-of-Flight cameras: Depth by correlation



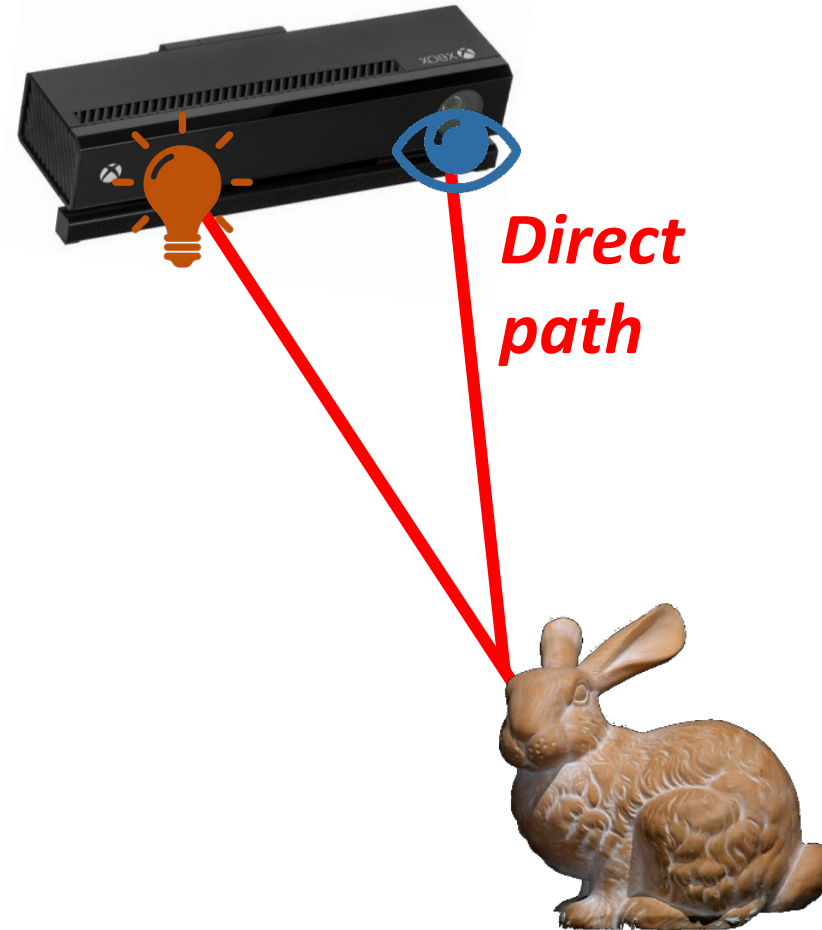
ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination

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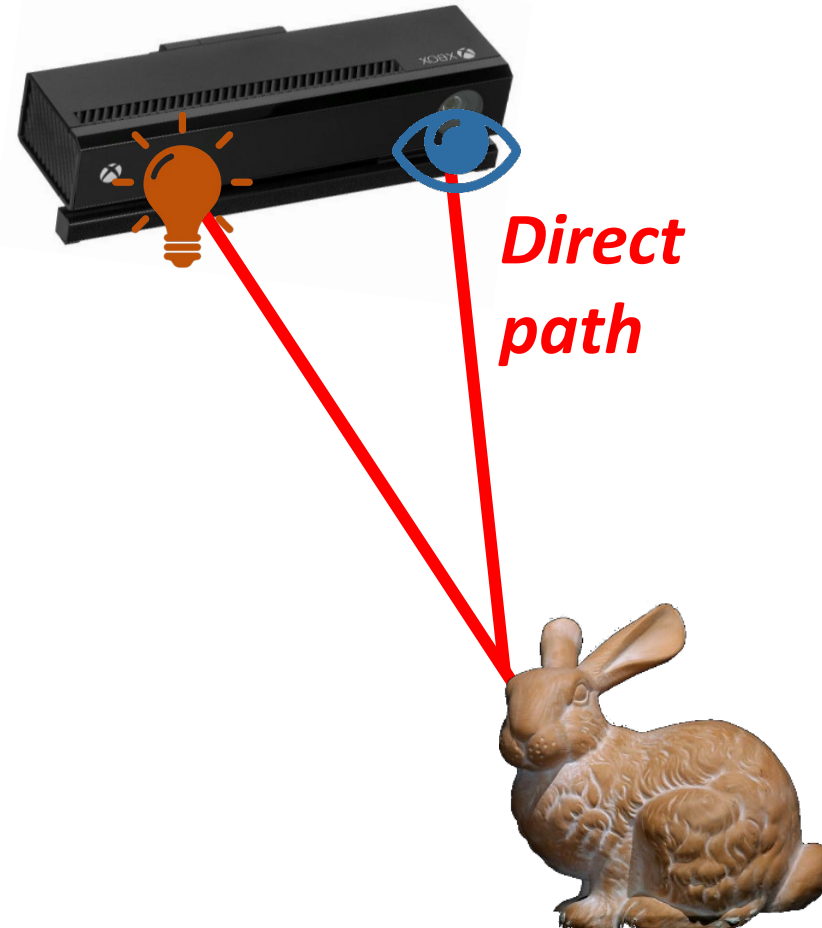
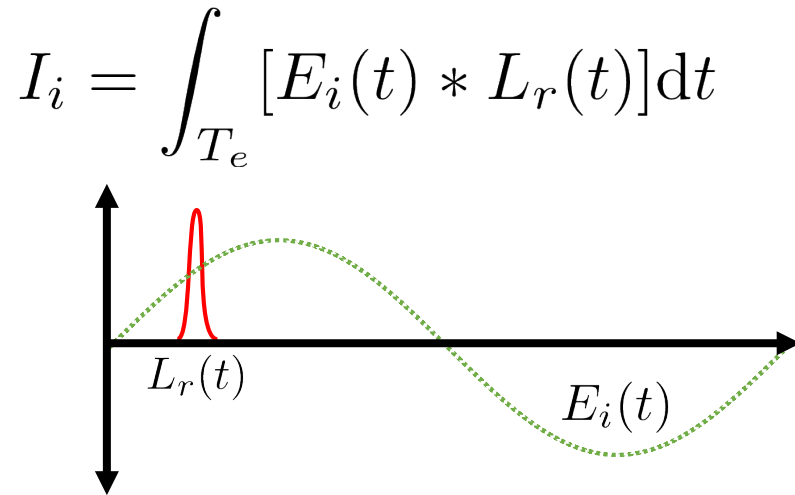
Assumes direct-only paths



ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination

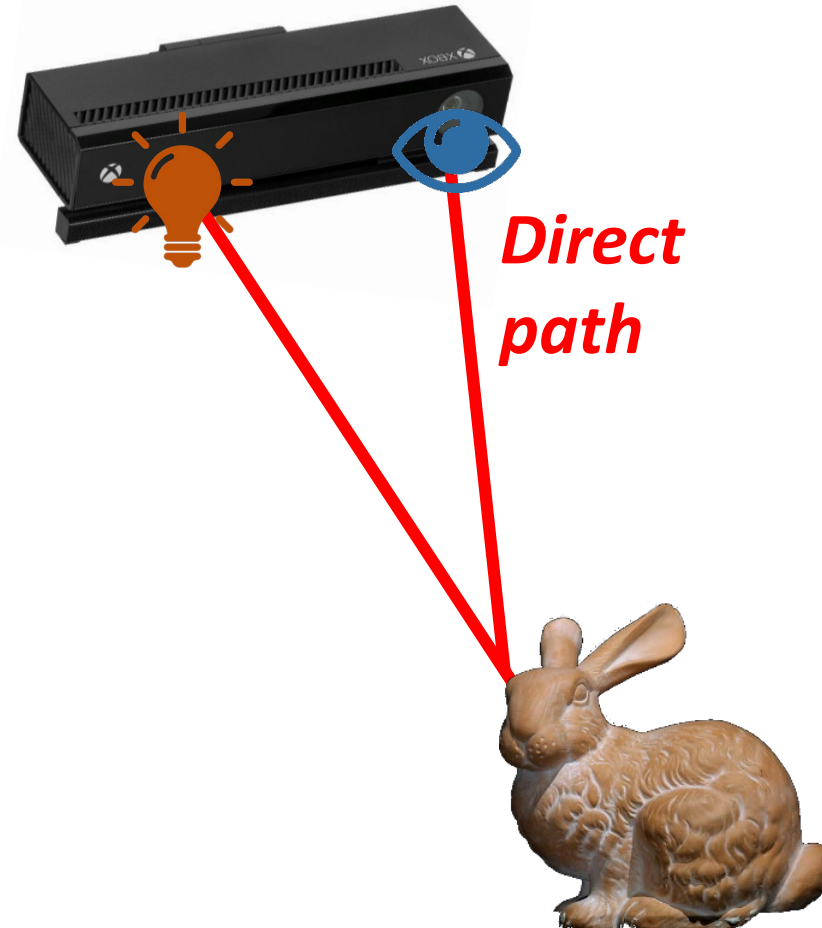
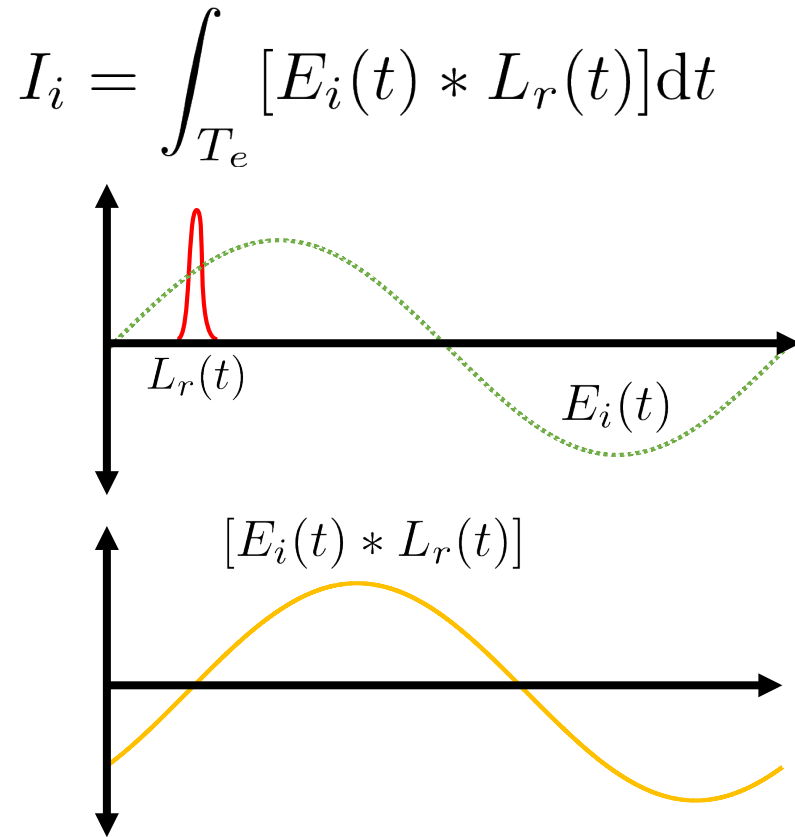
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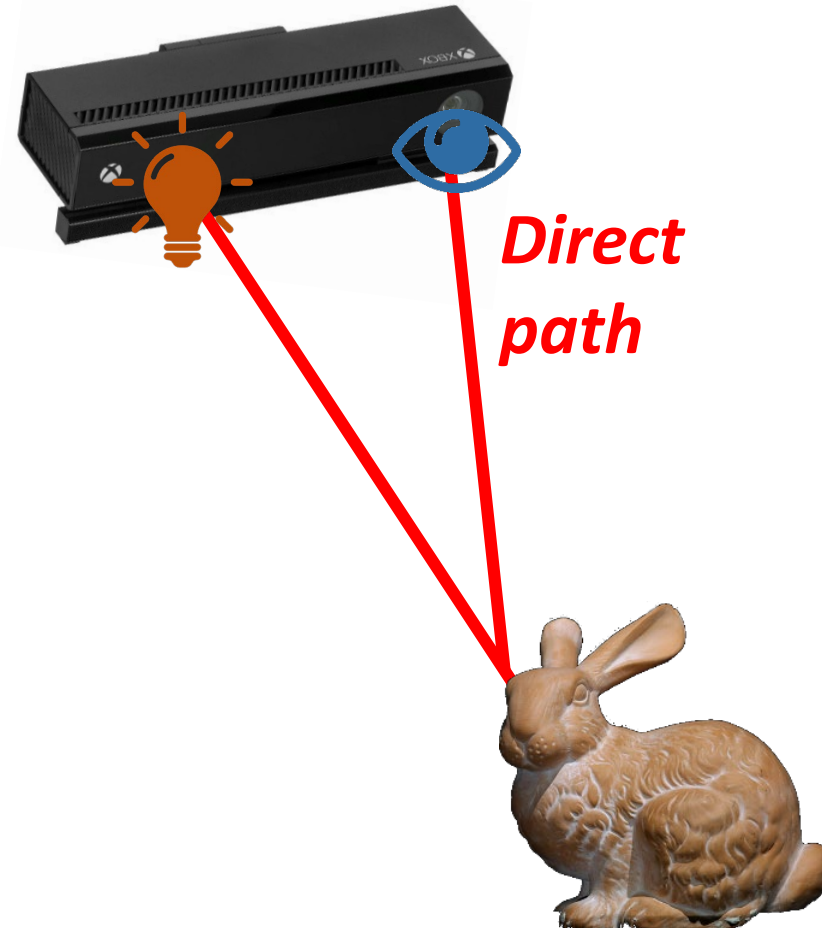
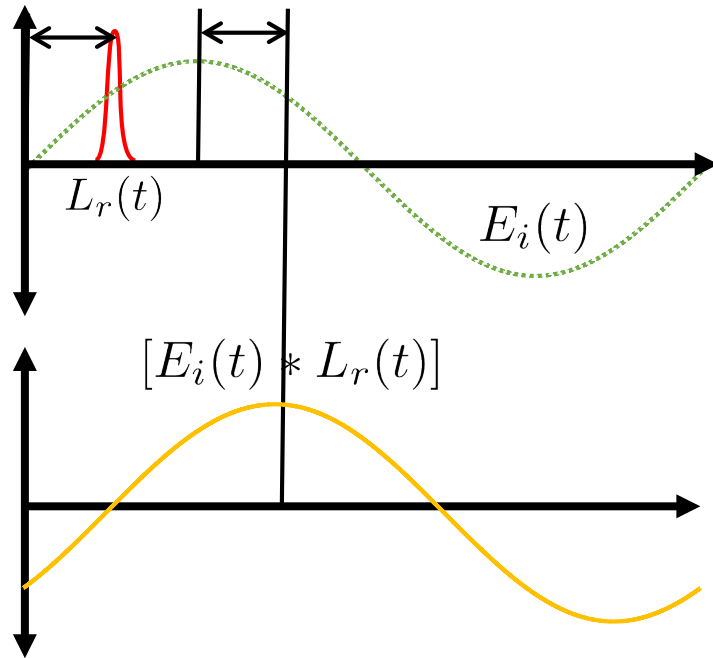


ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination

Assumes direct-only paths

$$I_i = \int_{T_e} [E_i(t) * L_r(t)] dt$$

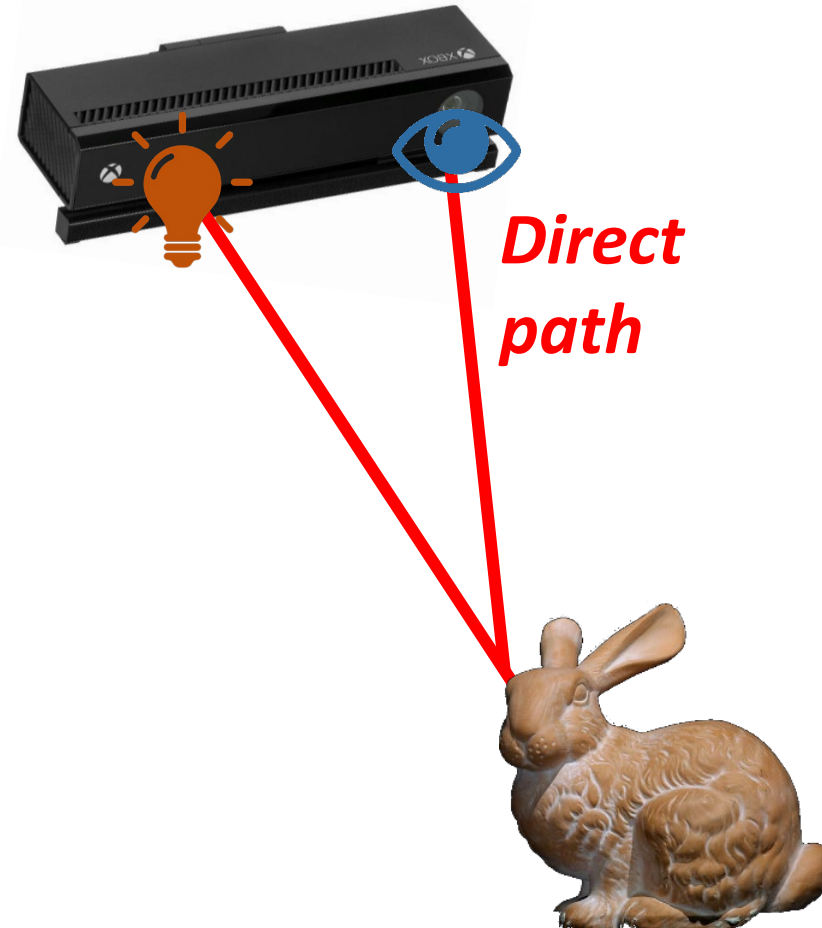
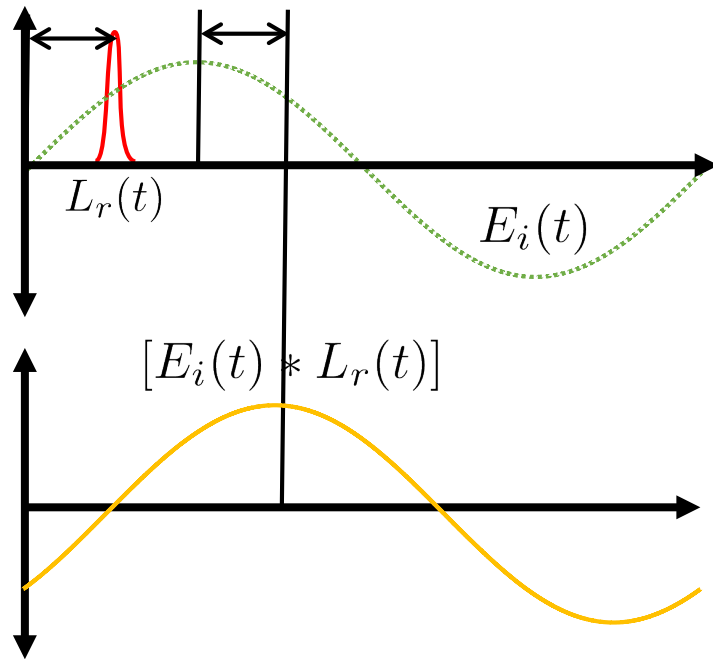


ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination

Assumes direct-only paths, but...

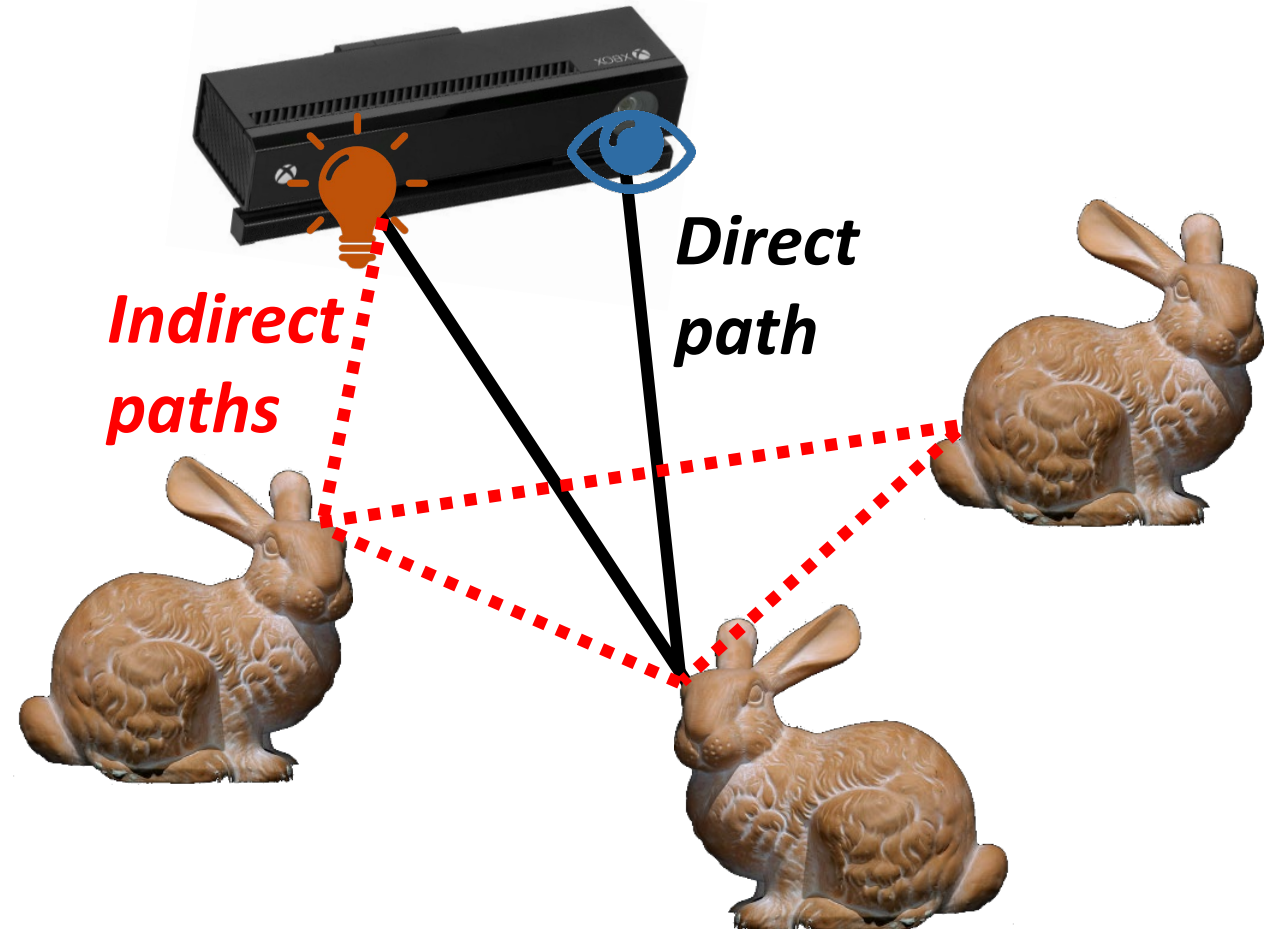
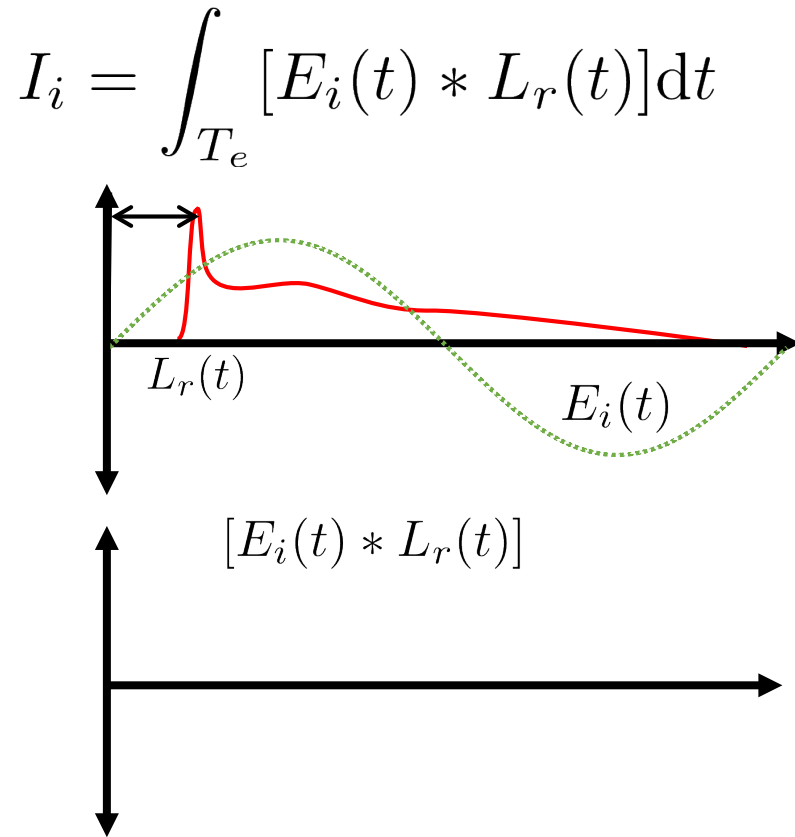
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ToF cameras: Challenges

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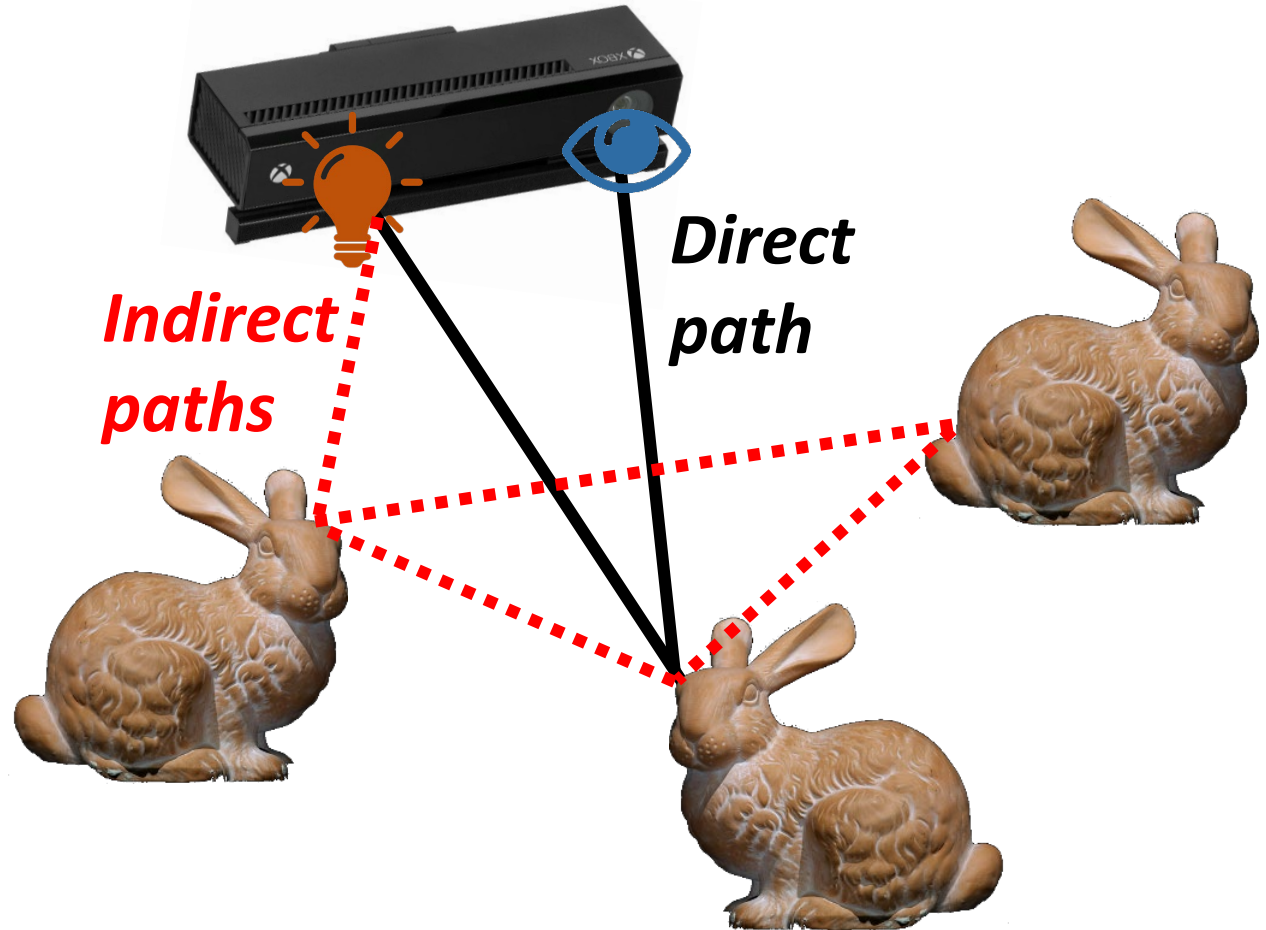
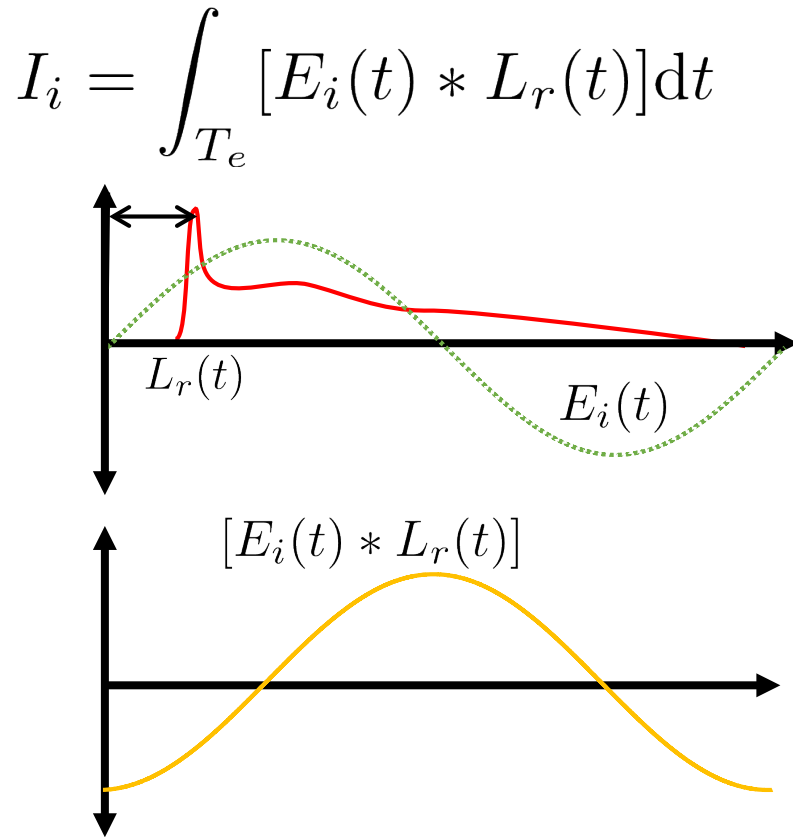
Indirect light → Longer paths



ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination

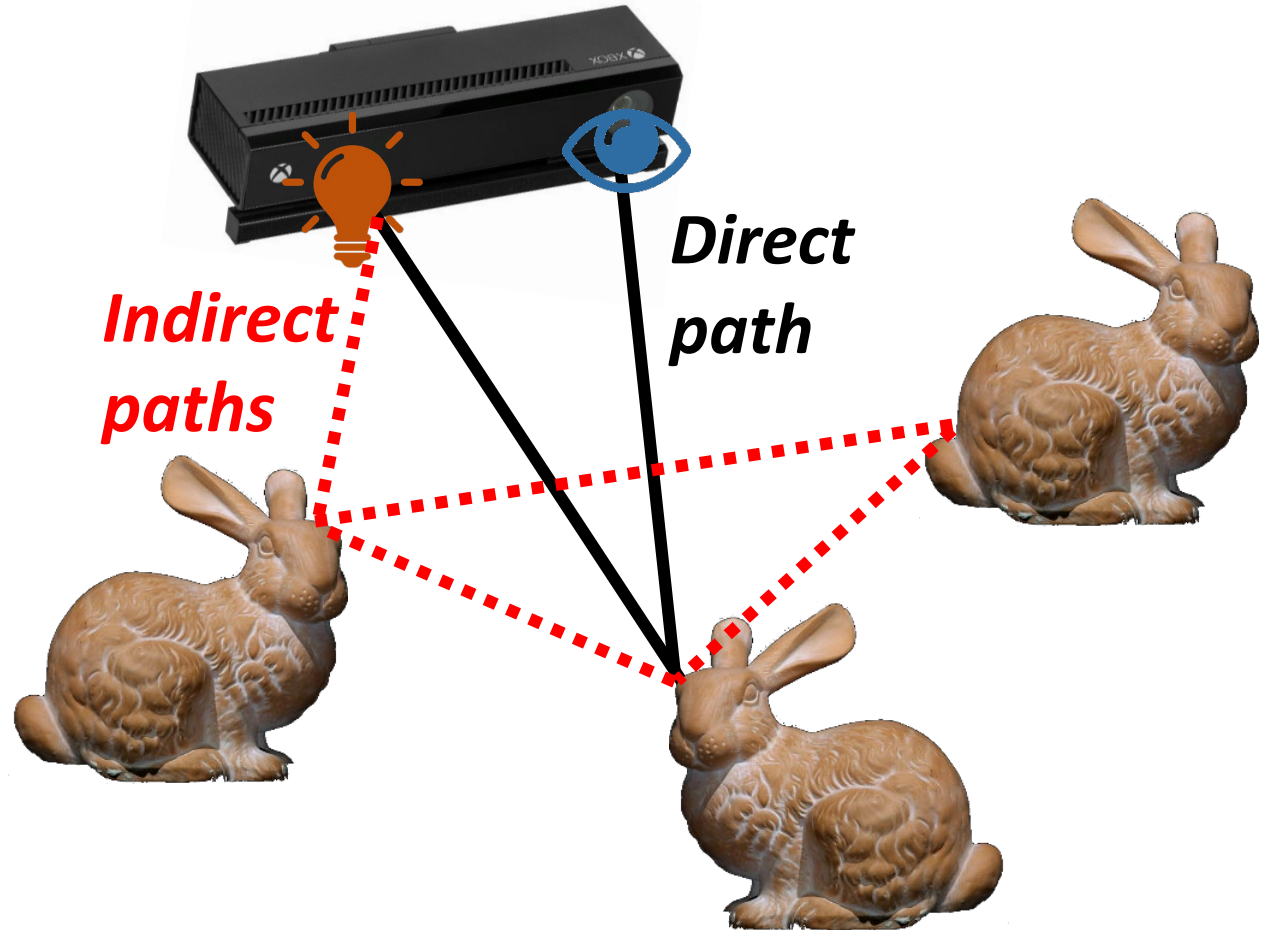
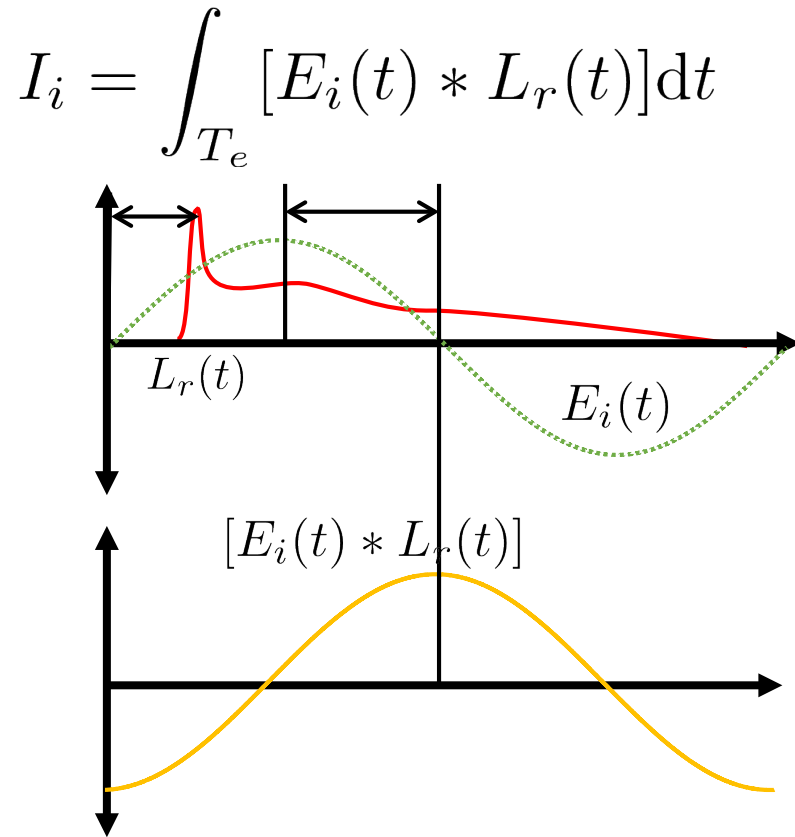
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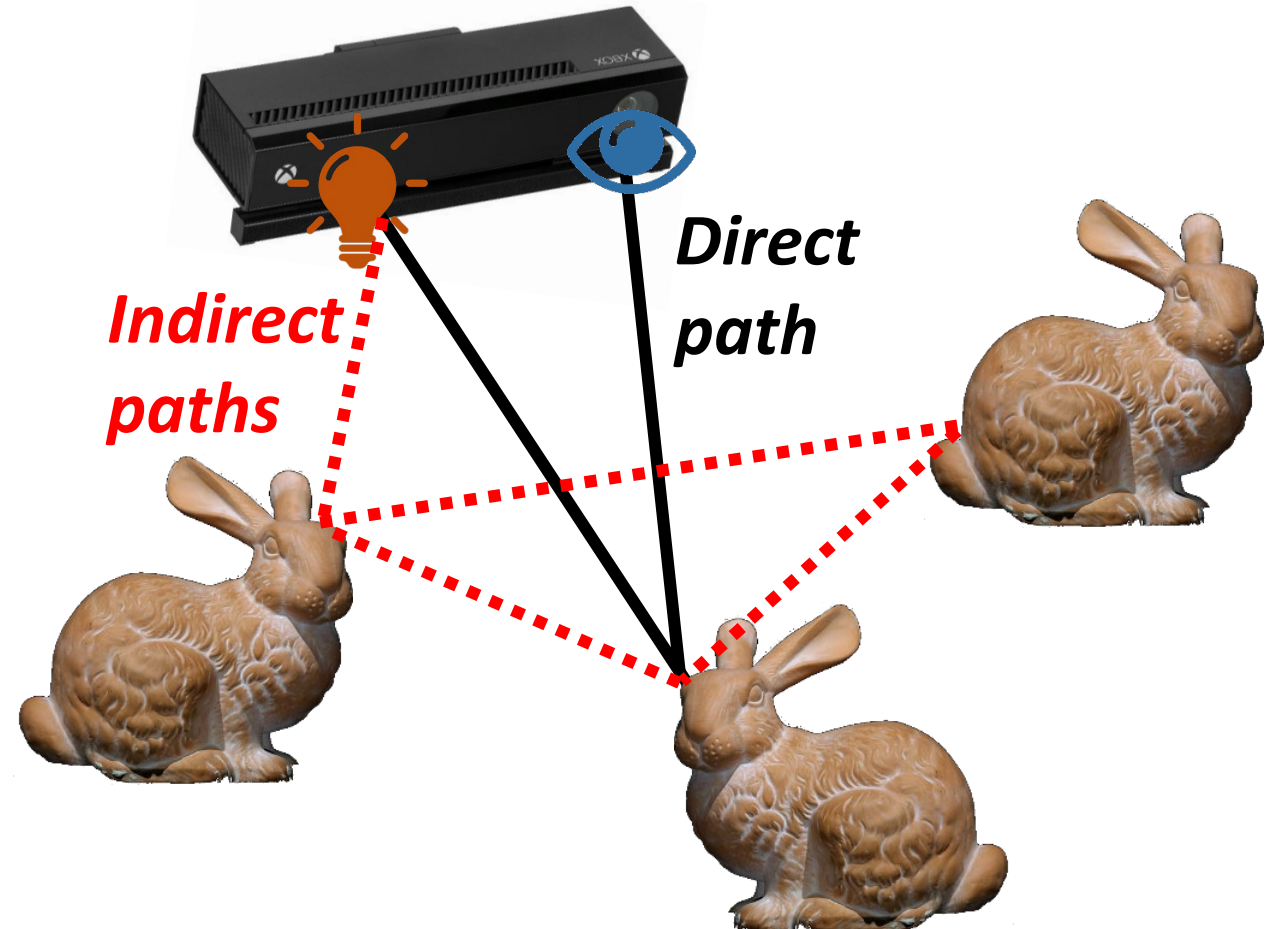
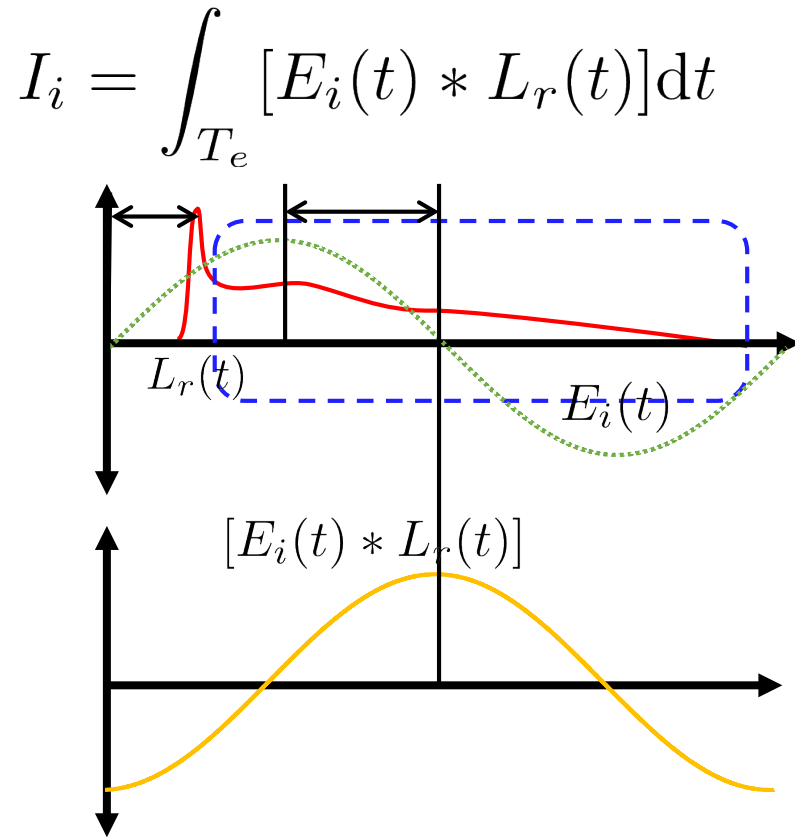
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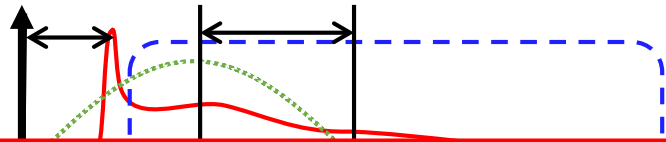


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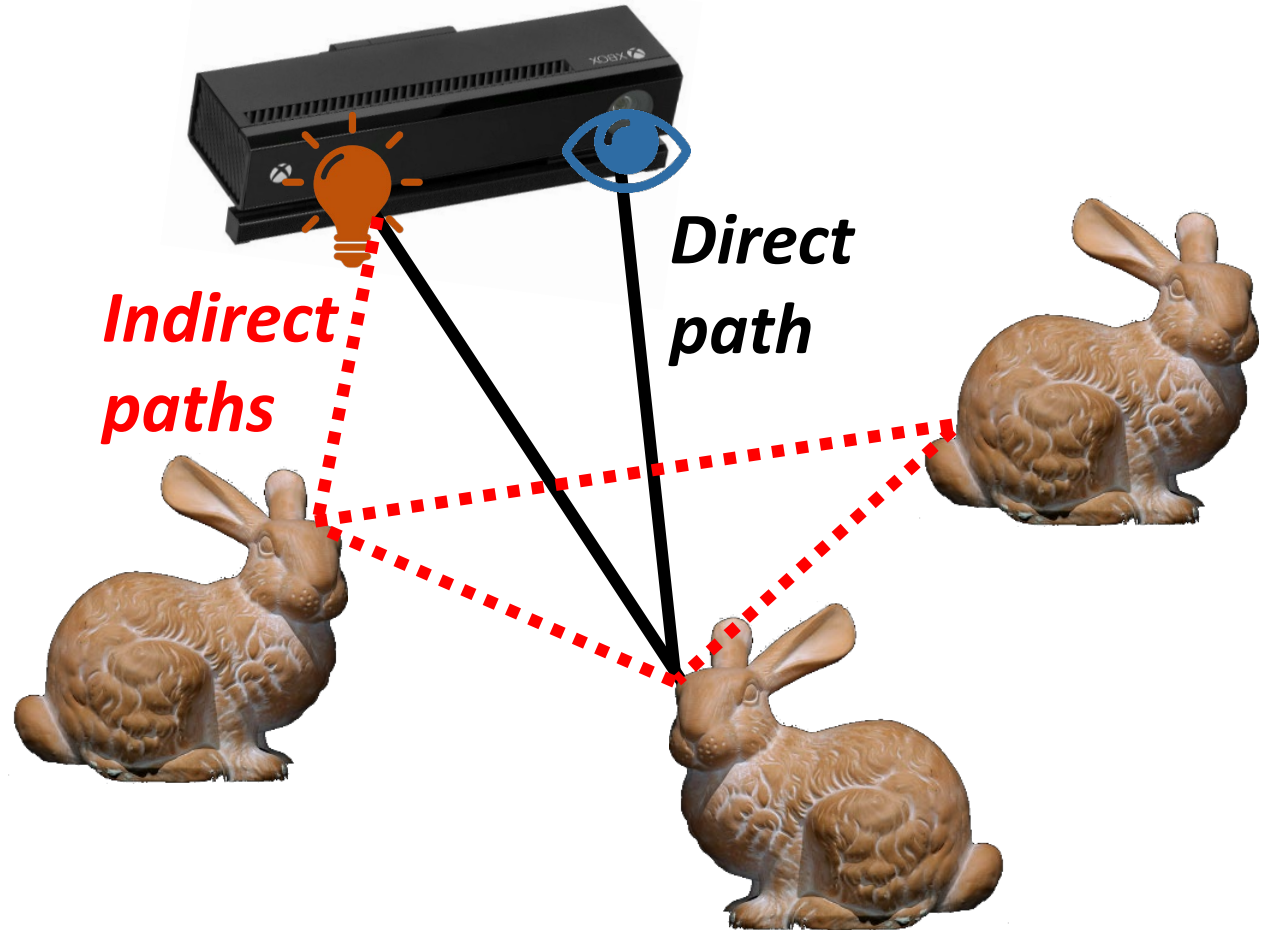
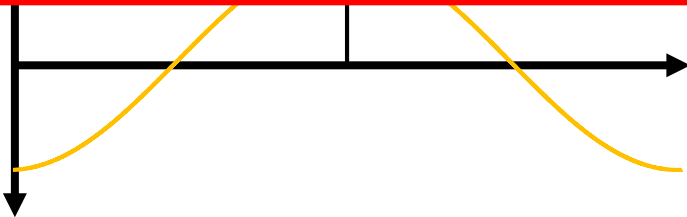
- Multipath interference: Depth errors due to global illumination

Indirect light → Longer paths

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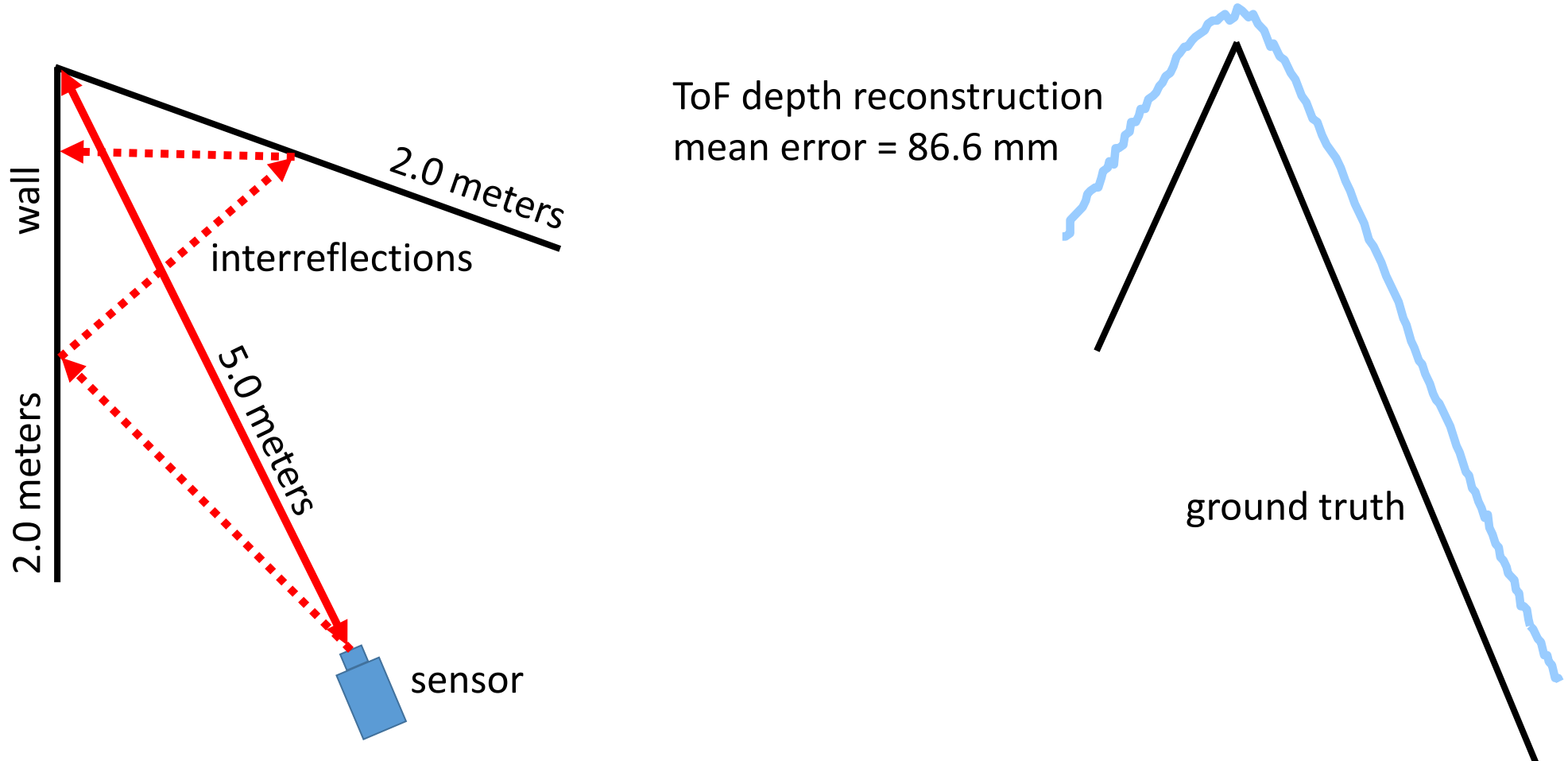


MULTIPATH INTERFERENCE
→ Depth overestimation



ToF cameras: Multipath Interference (MPI)

- Multipath interference: Depth errors due to global illumination



ToF cameras: Multipath Interference (MPI)

- **Multipath interference: Depth errors due to global illumination**



Synthetic dataset from [Marco et al. 2017]

Simulated benchmarks show **on average +12% depth overestimation** in general scenes.

ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination
- Translucent objects: Multiple depths on a single pixel

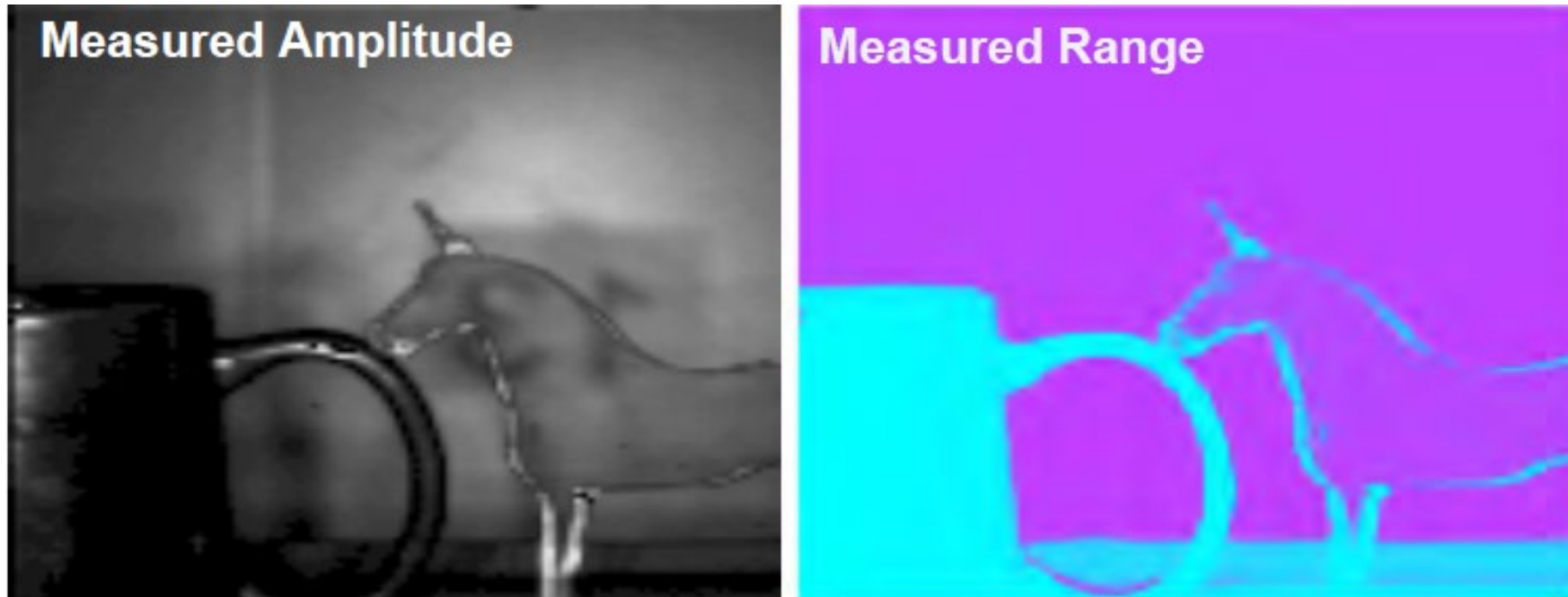


Image from [\[Kadambi et al. 2010\]](#)

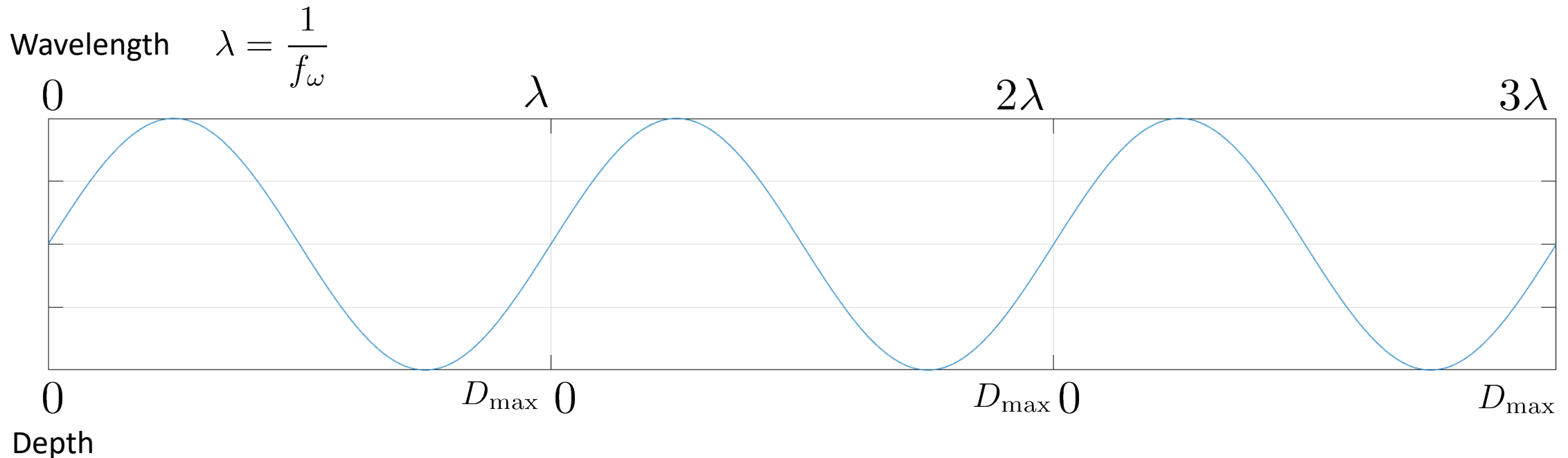
ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination
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- **Maximum depth ambiguity: Wavelength → Phase wrapping**

$$D_{\max} = \frac{c}{2f_{\omega}}$$

(speed of light)
(modulation frequency)



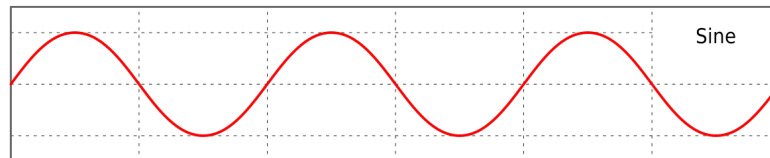
ToF cameras: Challenges

- Multipath interference: Depth errors due to global illumination
- Translucent objects: Multiple depths on a single pixel
- Maximum depth ambiguity: Wavelength $\rightarrow$ Phase wrapping
- Non-sinusoidal modulation: *Wiggling errors*

Arctan method assumes sinusoidal modulation

Square modulation is cheaper

Accurate



Cheaper

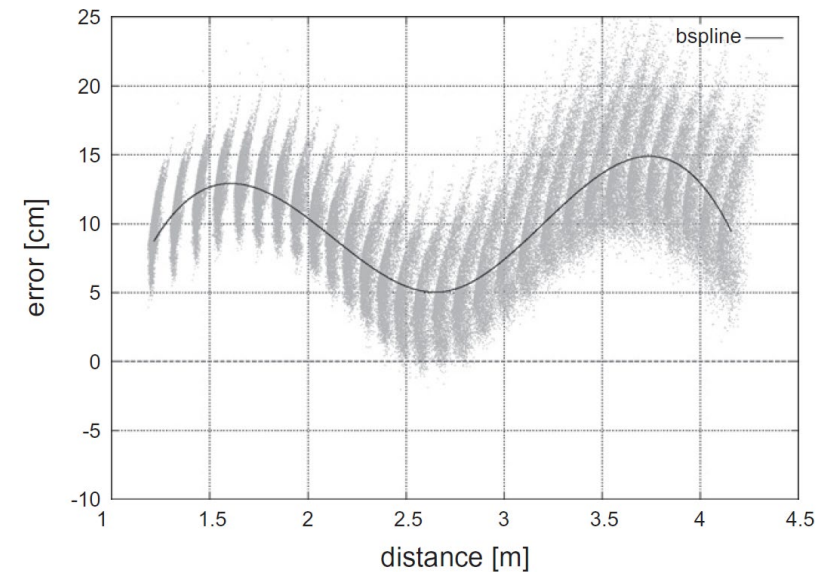
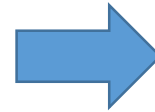
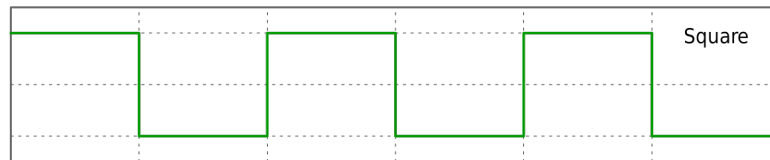


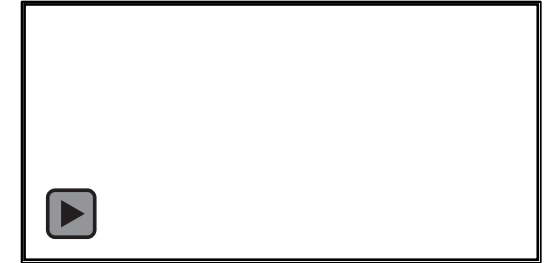
Image from [\[Lindner et al. 2010\]](#)

Transient imaging: Next week

Information hidden in the **temporal domain of light propagation**:

- Light in motion → Femtophotography
- Geometry reconstruction → Depth imaging
- **Seeing hidden objects → NLOS imaging**
 - Backprojection
 - Light cone transform
 - Fermat paths
 - Phasor Fields

Light in motion



NLOS imaging

